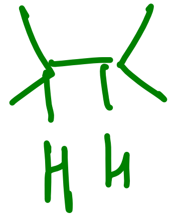
ORGANIC REACTION MECHANISMS

WHAT IS A CHEMICAL REACTION?

- A chemical reaction involves making and breaking bonds, with the gain, loss, or transformation of a fgp.

$$\rangle C=C\langle \; + \; H_2 \longrightarrow \text{No reaction}$$

$$\rangle C=C\langle \; + \; H_2 \xrightarrow[\text{Pd/Ni}]{\text{catalyst}} \text{Fast reaction}$$



$$C_6H_{12}O_6 + O_2 \longrightarrow \text{no reaction}$$

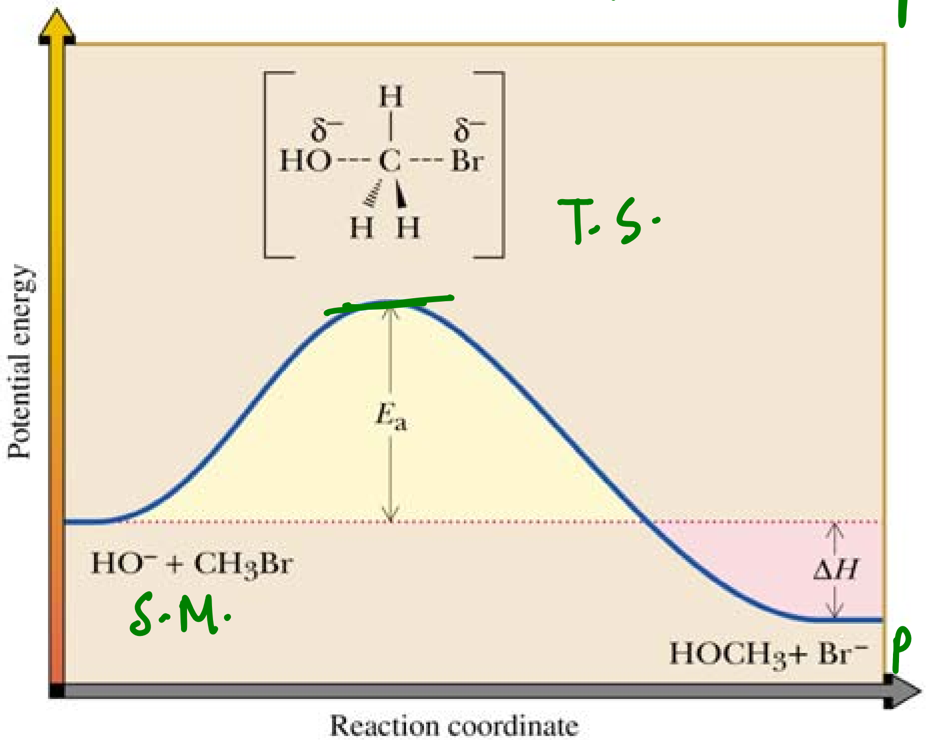
$$\xrightarrow{\text{enzymes}} CO_2 + H_2O$$

WHAT IS A REACTION MECHANISM ?

- It is a step by step map that shows the formation of products from starting materials.
- Which bonds are broken and which new ones are formed
- order & relative rates of bond-breaking and bond-forming steps
- Role of solvent & catalyst

ENERGY DIAGRAMS:

S_N2 : one step : concerted process



E_a : energy difference between reactants & transition state

$\Delta H_{format^n} = \Delta H_{diss\,(P)} - \Delta H_{diss\,(R)}$

$\Delta G^\circ = \underline{\Delta H^\circ} - \underline{T\Delta S^\circ}$ $\quad \Delta H < T\Delta S$

↓

$\Delta G^\circ < 0$: spontaneous

$\Delta G^\circ > 0$: non-spontaneous.

$H_P > H_R$: endothermic

heat is absorbed

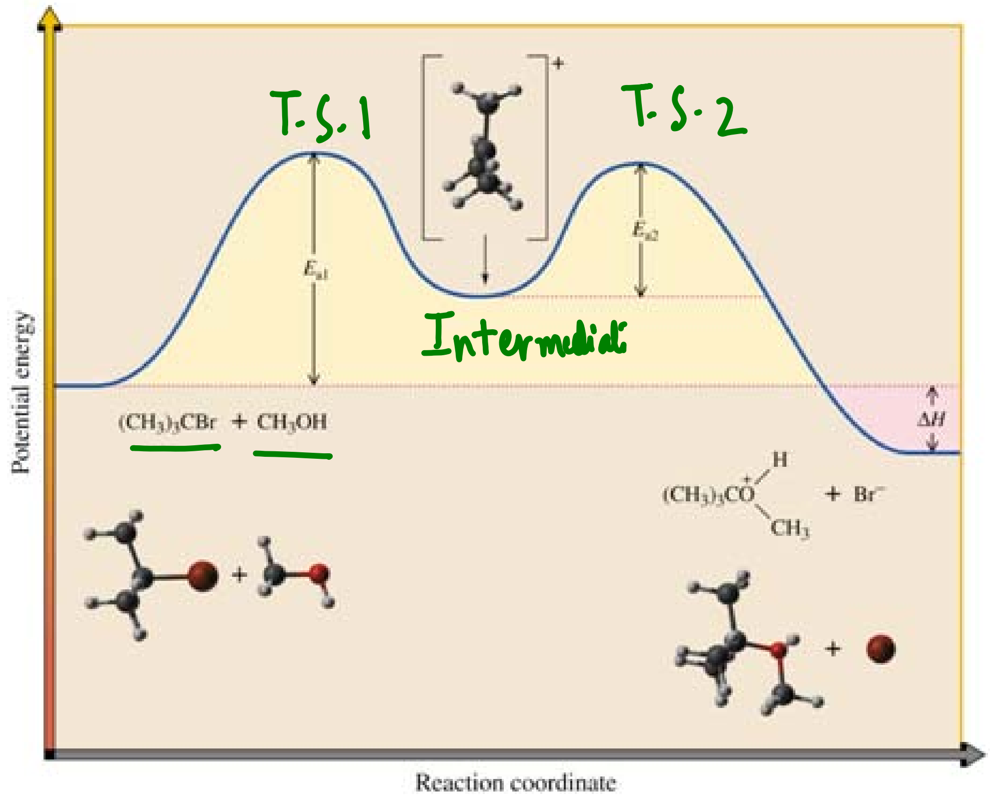
$\underline{H_P < H_R}$: exothermic

heat is released

TRANSITION STATE:

- An unstable species of maximum energy formed during the course of reaction.
- It has a definite geometry, arrangement of bonding & non-bonding electrons.
- Cannot be isolated & structure cannot be determined experimentally.
- Lifetime is less than one picosecond. (duration of a single bond vibration)

2 Step Process : Non-concerted



- A species formed between two successive reaction steps that lies in the energy minimum between transition states
- Can be isolated
- Most common intermediates carbocations, carbanions, free radicals.

S_N1

Kinetic versus Thermodynamic Controlled Reactions

1. DAR : cyclo addition → DIELS - ALDER REACTION
2. Conjugated diene addition
3. Enolate alkylation
4. Substitution reaction of naphthalene

Thermodynamic factor

It is related to stability of the products

It describes the properties of a system at equilibrium

Keq = [products] / [reactants]

It is only concerned with the final state of a system and the mechanism of transformation does not matter at all.

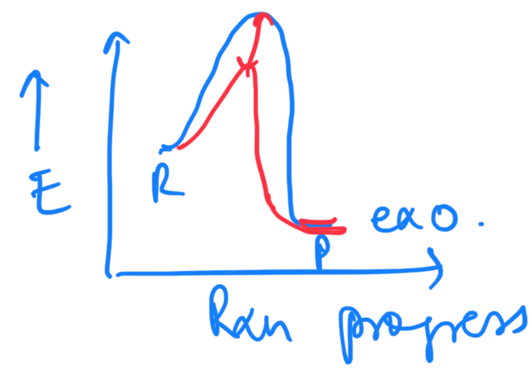
Kinetic factor

It is related to the reactivity of the reactants

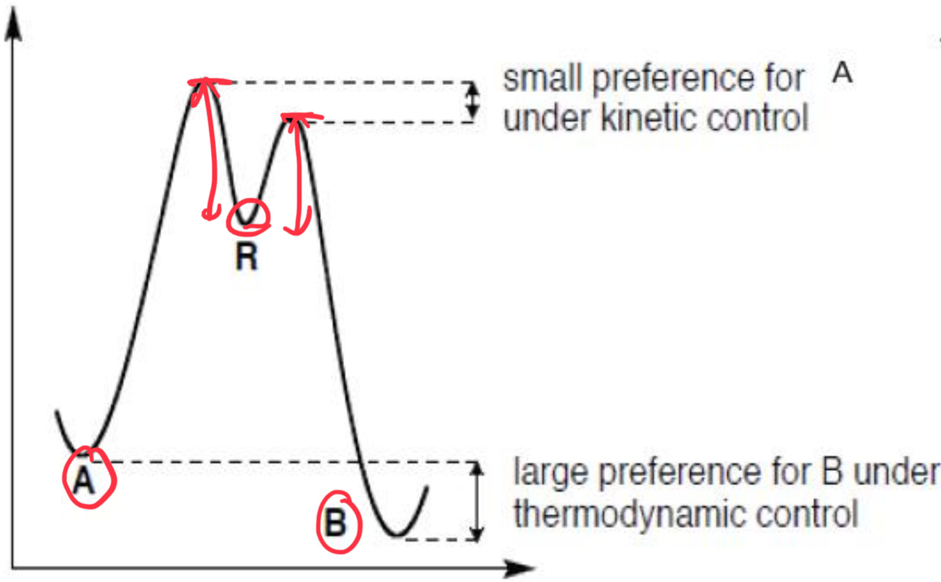
It describes the rates of product formation, and how fast equilibrium is reached

Rate of reaction = (number of collisions per unit time) **x** (fraction with sufficient energy) **x** (fraction with proper orientation)

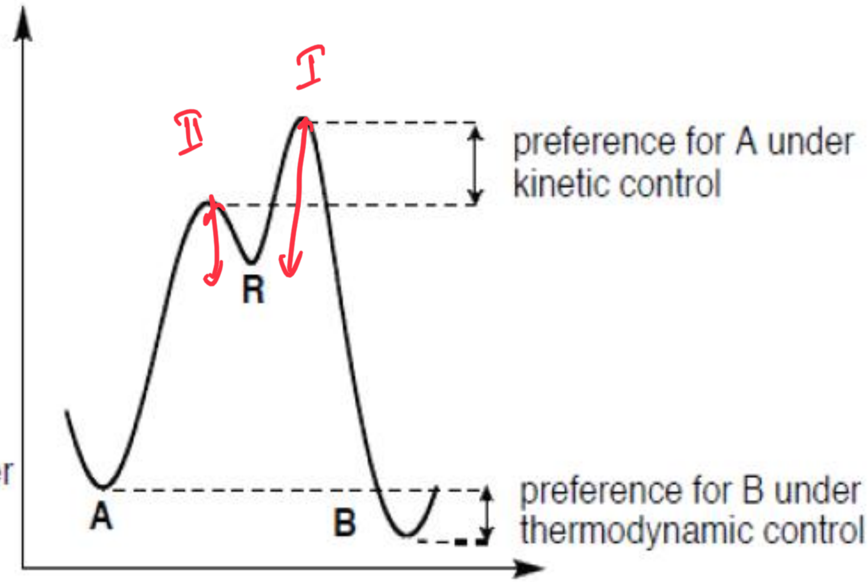
Kinetic stability is largely dependent on the activation energy for the reaction, and the pathway between reactants and products



Case 1: **B** is both the kinetic and thermodynamic product



Case 2: **A** is the kinetic product while **B** is the thermodynamic product



Energy profile for a thermodynamic and kinetic product

Thermodynamic control

Involves higher temperatures and long reaction times to ensure that even the slower reactions have a chance to occur, and all the material is converted to the most stable compound.

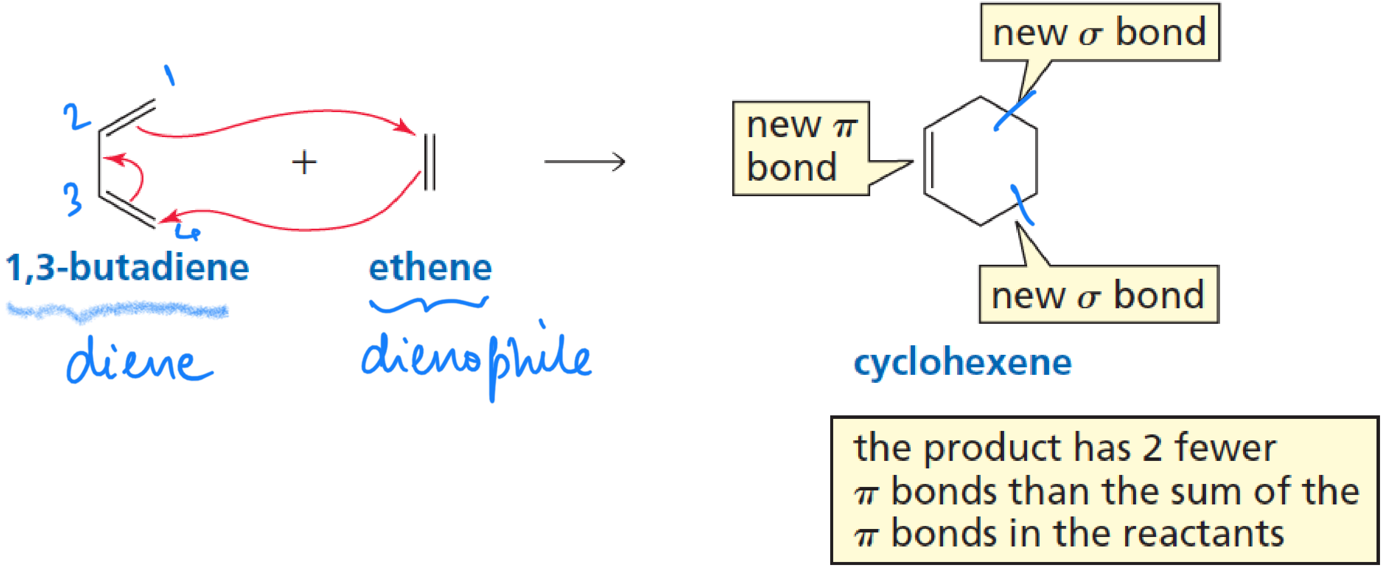
The thermodynamic product predominates when the reaction is reversible (thermodynamic control)

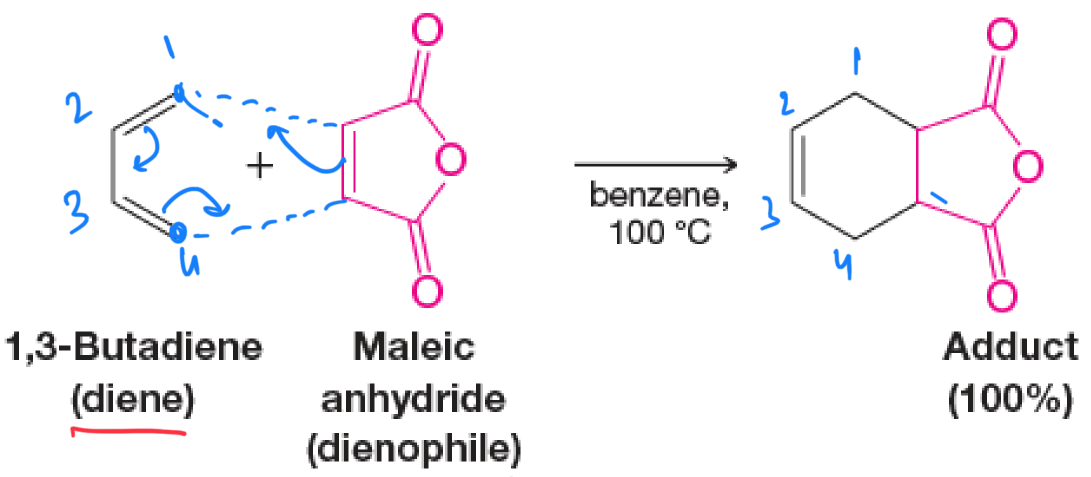
Kinetic control

A kinetically controlled reaction involves lower temperatures and shorter reaction times, which ensures that only the fastest reaction has the chance to occur

The kinetic product predominates when the reaction is irreversible (kinetic control)

a cycloaddition reaction

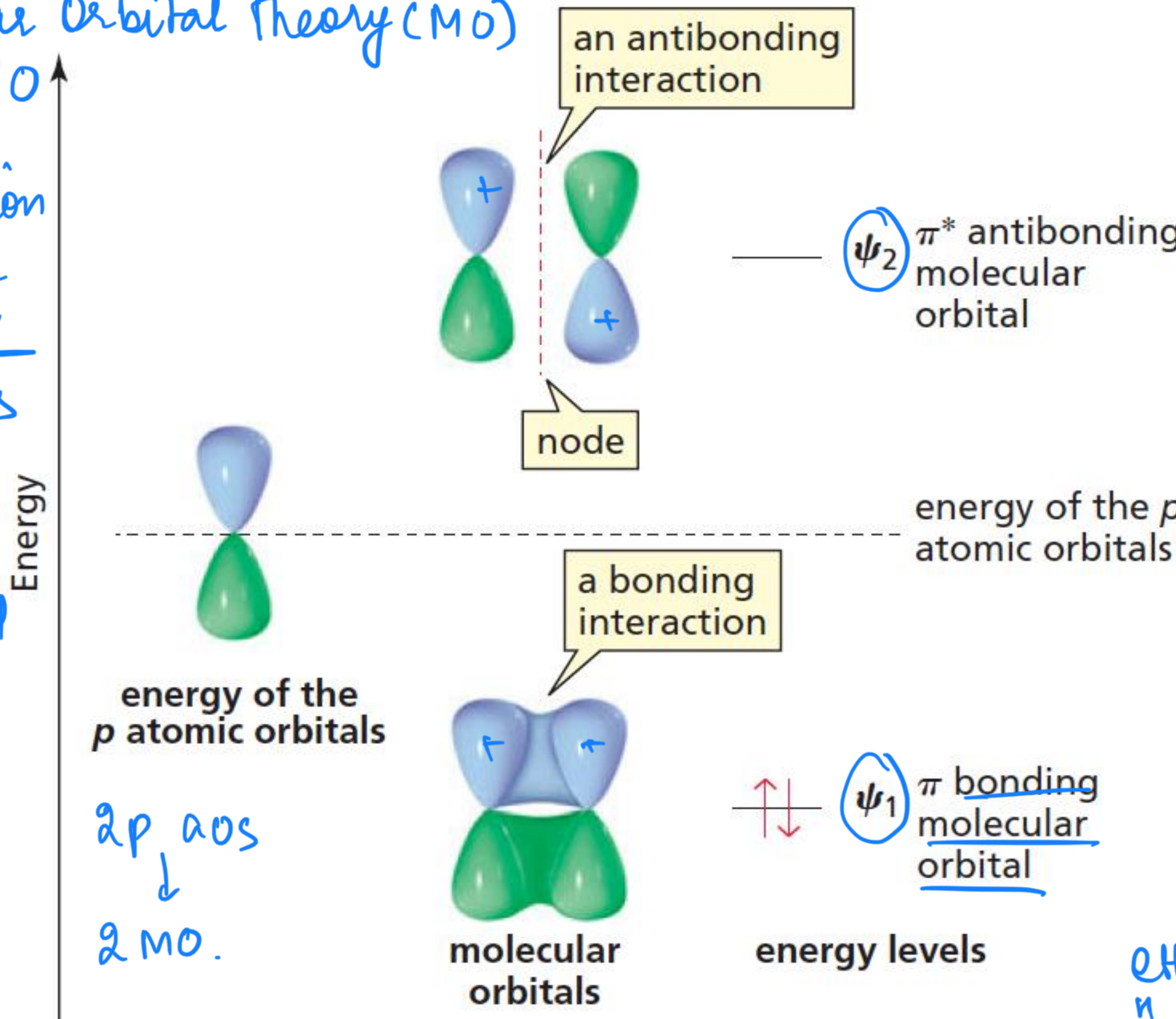




Molecular Orbital Theory (MO)

LCAO = MO
Linear Combination of atomic orbitals.

No. of AOs combine = No. of MOs generated



subtractive

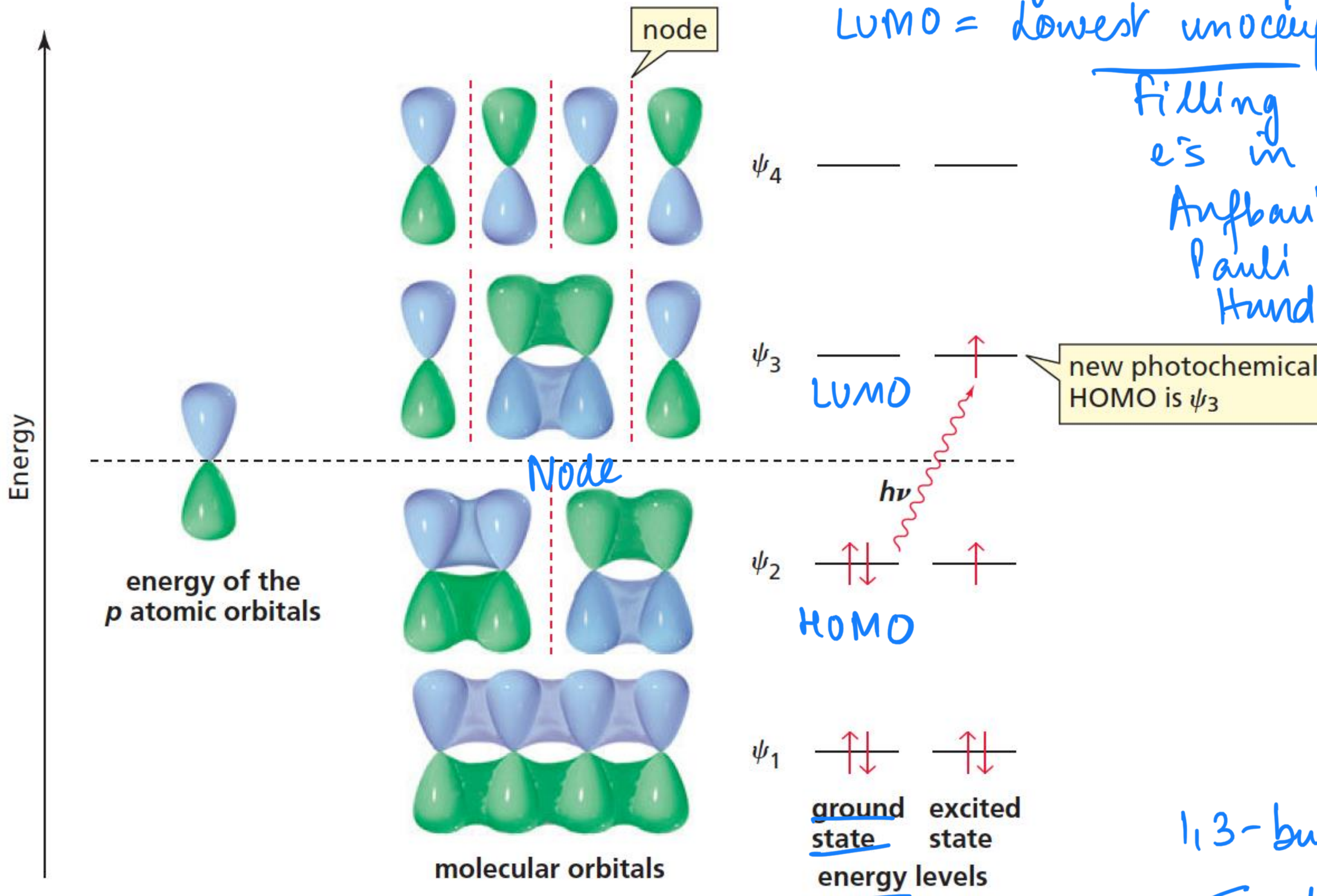
additive

2p aos
↓
2 MO.

ethene

HOMO = Highest occupied MO
LUMO = Lowest unoccupied MO.

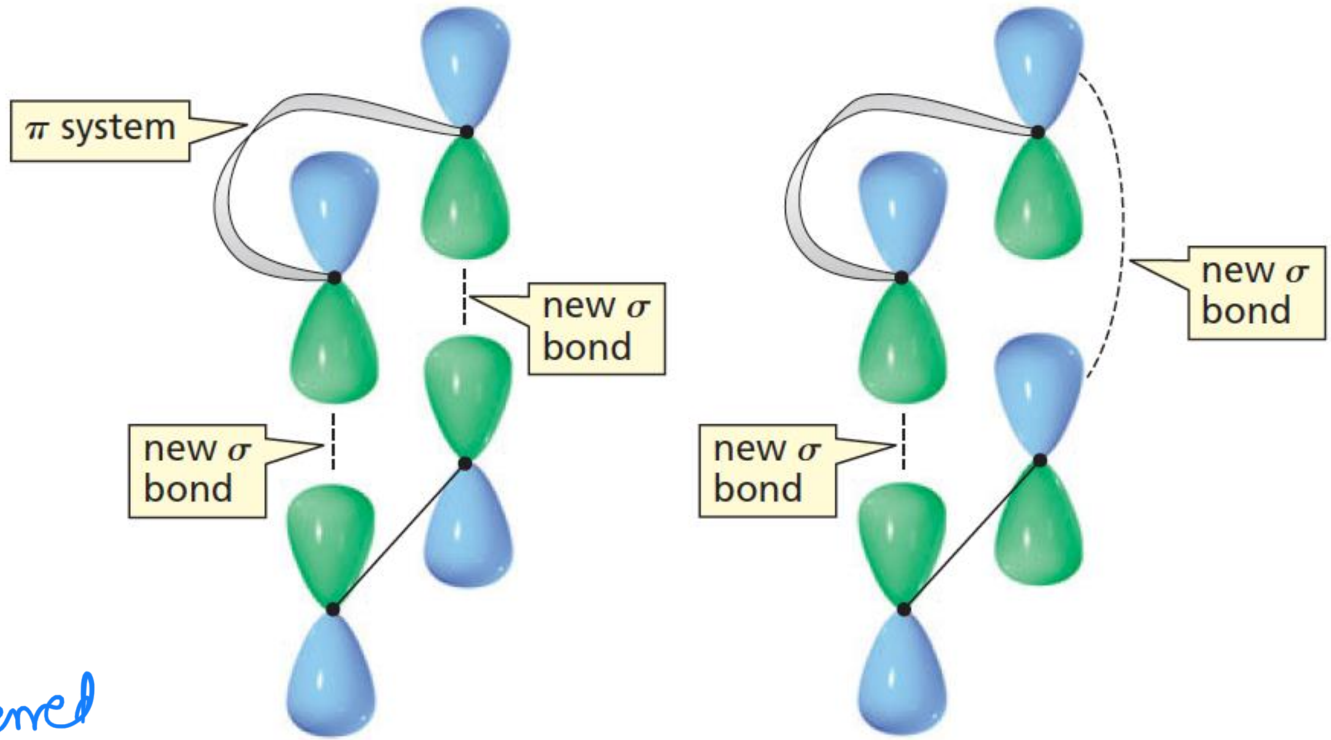
filling up of e's in π MOs –
Aufbau's
Pauli
Hund's



1,3-butadiene

E 4p aos
$\downarrow$ LCAO
4π MOs

DAR: Conservation of orbital symmetry



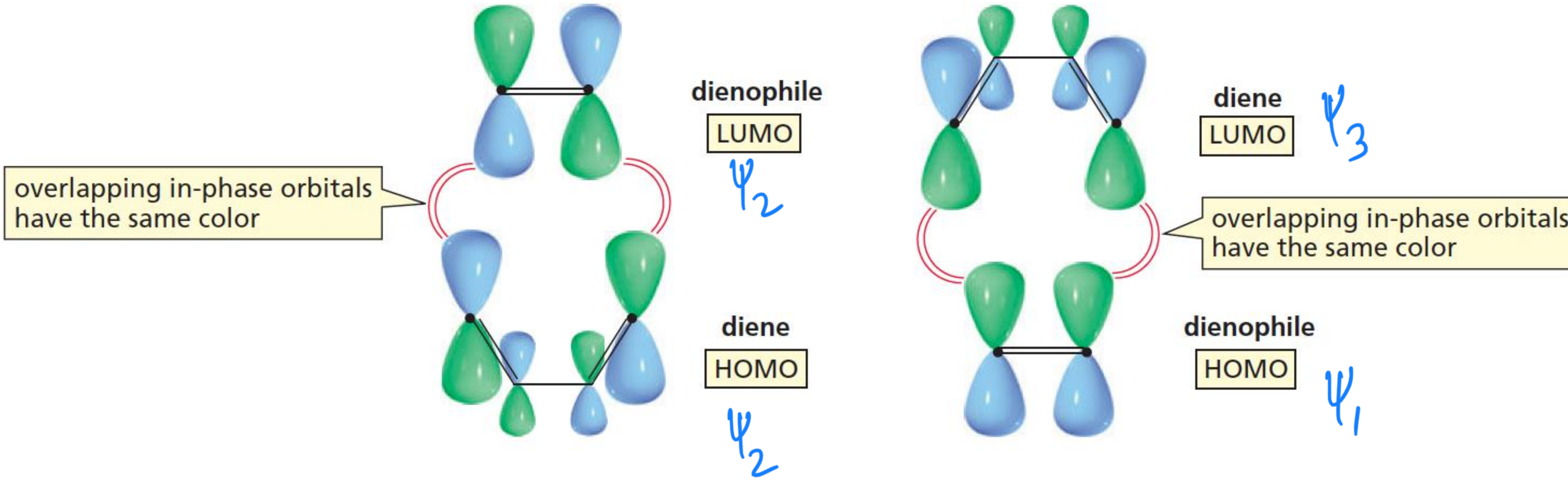
suprafacial bond formation

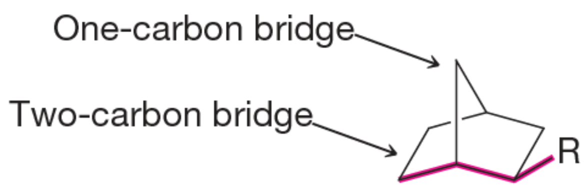
antarafacial bond formation

preferred lower rings upto 6C →

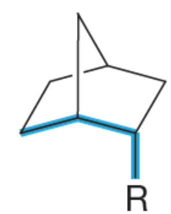
Symmetry is conserved

DAR HOMO + LUMO of the two π system containing reactants





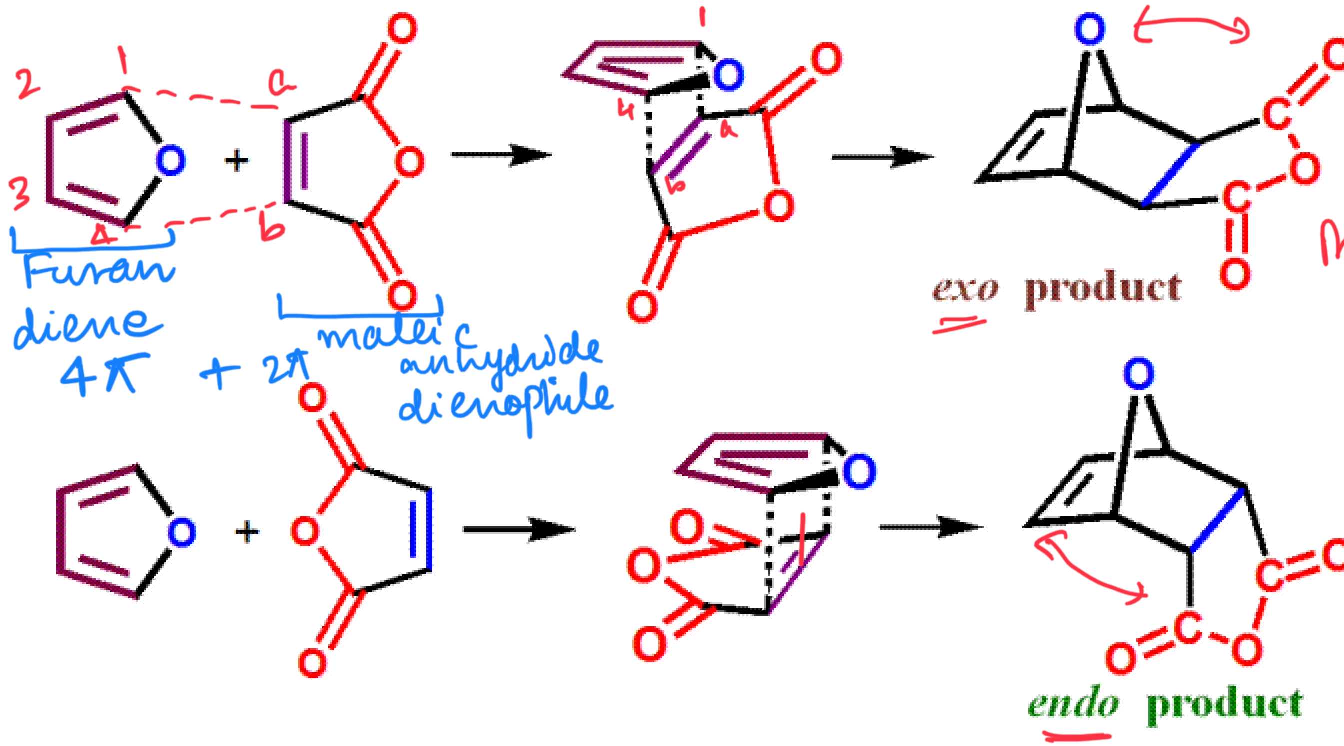
R is exo
(oriented away from the longest bridge)



R is endo
(oriented closer to the longest bridge)

Exo approach: the dienophile aligns with its electron-withdrawing substituent below and away from the diene π system, as shown, or above it

Kinetically controlled
vs
Thermodynamically controlled
DAR



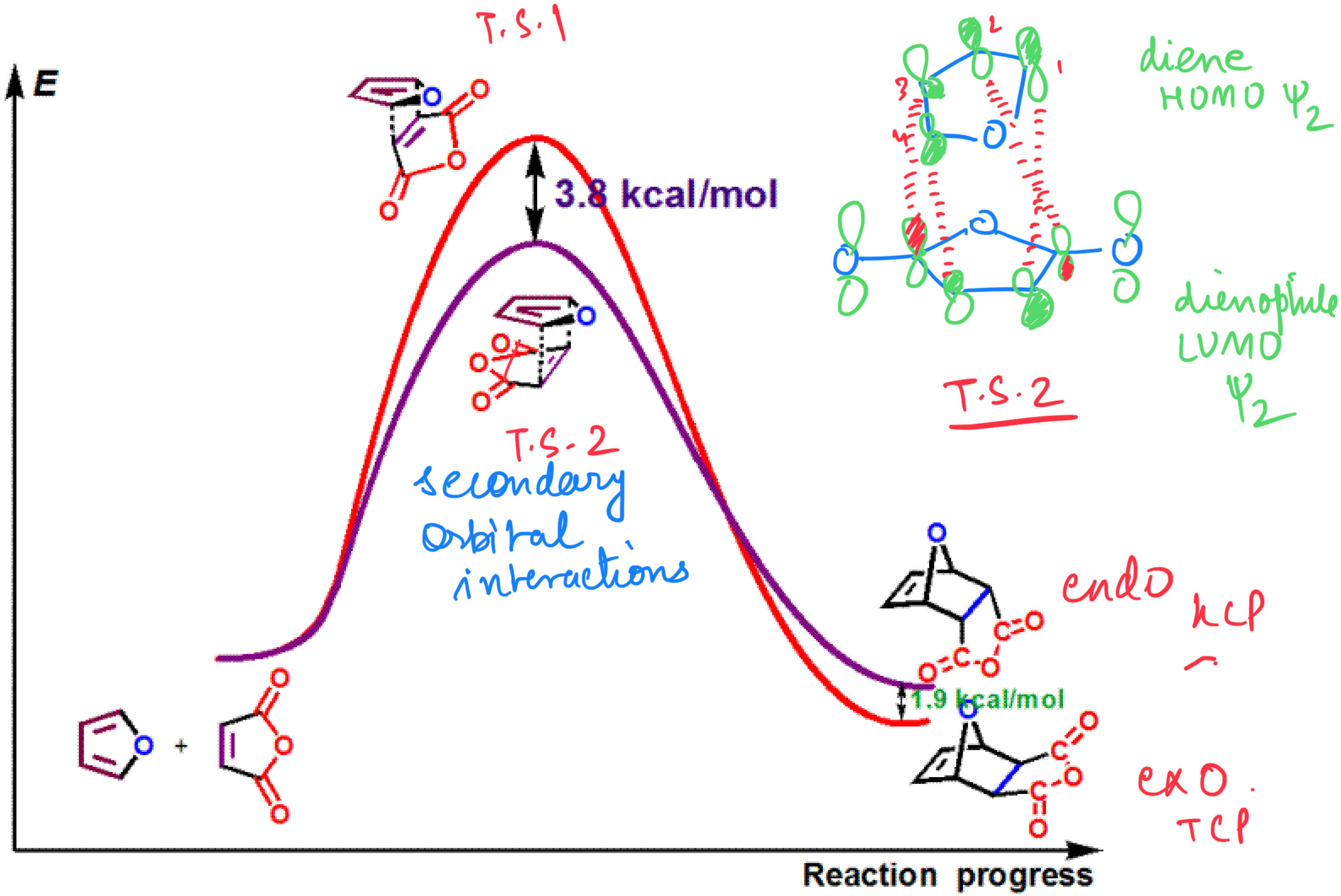
thermodynamic

Kinetic product

Experiment: *exo* product more stable by 1.9 kcal/mol TCP

E_a lower for the *endo* product by 3.8 kcal/mol · KCP

Endo approach: the dienophile aligns with its electron-withdrawing substituent underneath the π system, as shown, or directly above it


E
T.S.1
3.8 kcal/mol
T.S.2
secondary orbital interactions
diene HOMO ψ_2
dienophile LUMO ψ_2
T.S.2
endo
kcp
1.9 kcal/mol
exo
TCP
Reaction progress